MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PRE-BOARD MOCK EXAMINATION – 12** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

MATHS MANTHAN | $6 \times 1 = 6$ Marks

Q1. The value of $\sqrt{6+\sqrt{6+\sqrt{6+\ldots}}}$ is: [1]

(A) 4 **(B)** 3
(C) -2 **(D)** 3.5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 3

Let $x = \sqrt{6+\sqrt{6+\sqrt{6+\ldots}}}$. Clearly $x > 0$ (principal square root).

Since the nested radical repeats indefinitely, we can write $x = \sqrt{6+x}$.

Formula Used: $x = \sqrt{6+x} \implies x^2 = 6+x \quad (x>0)$

Rearranging and factorising:

$$x^2 - x - 6 = 0 \implies (x-3)(x+2) = 0 \implies x = 3 \text{ or } x = -2$$

Since $x > 0$, we reject $x = -2$. Thus $\boxed{x = 3}$. Hence, option **(B)** is correct. **[1 mark]**

Q2. If the quadratic equation $x^2 - kx + 4 = 0$ has no real roots, then: [1]

(A) $-4 < k < 4$ **(B)** $k > 4$
(C) $k < -4$ **(D)** $k = \pm 4$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $-4 < k < 4$

Comparing $x^2 - kx + 4 = 0$ with $ax^2 + bx + c = 0$, we have $a = 1$, $b = -k$, and $c = 4$.

Formula Used: No real roots $\iff D = b^2 - 4ac < 0$

$$(-k)^2 - 4(1)(4) < 0 \implies k^2 - 16 < 0 \implies k^2 < 16 \implies \boxed{-4 < k < 4}$$

Hence, option **(A)** is correct. **[1 mark]**

Q3. If the sum and product of the roots of the equation $kx^2 + 6x + 4k = 0$ are equal, then $k =$: [1]

(A) $-\frac{3}{2}$ **(B)** $\frac{3}{2}$

(C) $\frac{2}{3}$ **(D)** $-\frac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $-\frac{3}{2}$

For $kx^2 + 6x + 4k = 0$ $(k \neq 0)$, we have $a = k$, $b = 6$, and $c = 4k$.

Formula Used: Sum $(\alpha + \beta) = -\frac{b}{a}$, Product $(\alpha\beta) = \frac{c}{a}$

Given $\alpha + \beta = \alpha\beta$:

$$-\frac{6}{k} = \frac{4k}{k} \implies -\frac{6}{k} = 4 \implies 4k = -6 \implies k = -\frac{6}{4} = \boxed{-\frac{3}{2}}$$

Hence, option **(A)** is correct. **[1 mark]**

Q4. If one root of the quadratic equation $2x^2 + kx + 4 = 0$ is 2, then the other root is: [1]

(A) 6 **(B)** -6

(C) -1 **(D)** 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) 1

Let the roots of $2x^2 + kx + 4 = 0$ $(a = 2, c = 4)$ be $\alpha = 2$ and β.

Formula Used: Product of roots: $\alpha\beta = \frac{c}{a}$

$$2 \cdot \beta = \frac{4}{2} = 2 \implies \boxed{\beta = 1}$$

(Alternatively: $2(4) + 2k + 4 = 0 \implies k = -6 \implies 2x^2 - 6x + 4 = 2(x - 2)(x - 1) = 0$.) Hence, option **(D)** is correct. **[1 mark]**

Q5. If the roots of the quadratic equation $x^2 - px + q = 0$ differ by 1, then: [1]

(A) $p^2 = 4q + 1$ **(B)** $p^2 = 4q - 1$

(C) $q^2 = 4p + 1$ **(D)** $p^2 + 4q = 1$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $p^2 = 4q + 1$

Let the roots of $x^2 - px + q = 0$ be α and β with $\alpha - \beta = 1$ (or $|\alpha - \beta| = 1$).

Here $\alpha + \beta = p$ and $\alpha\beta = q$.

Formula Used: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

Substituting $|\alpha - \beta| = 1$, $\alpha + \beta = p$, and $\alpha\beta = q$:

$$(1)^2 = p^2 - 4q \implies \boxed{p^2 = 4q + 1}$$

Hence, option **(A)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): The quadratic equation $(x-2)^2=0$ has two equal real roots: $x=2,\ 2$.

Reason (R): A quadratic equation always possesses exactly two roots (which may be distinct, coincident, or non-real).

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Expanding $(x-2)^2=x^2-4x+4=0$ gives $a=1$, $b=-4$, and $c=4$.

Formula Used: $D=b^2-4ac=0 \implies$ two coincident (equal) real roots

Computing $D=(-4)^2-4(1)(4)=16-16=0$, and $(x-2)(x-2)=0$ yields two coincident real roots $x=2,2$.

Both Assertion (A) and Reason (R) are **true**, and Reason (R) explains why an equation whose linear factor is repeated counts as having two equal real roots rather than a single root. **[1 mark]**

Section B: Very Short Answer Questions

MATHS MANTHAN | $2\times 2=4$ Marks

Q7. Solve the following equation for x: $\dfrac{1}{x}-\dfrac{1}{x-2}=3, \quad x\neq 0,\ 2.$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Combining the fractions on the LHS over the common denominator $x(x-2)$ $(x\neq 0,2)$:

Formula Used: $\dfrac{1}{x}-\dfrac{1}{x-2}=\dfrac{(x-2)-x}{x(x-2)}, \qquad x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$

$$\frac{x-2-x}{x(x-2)}=3 \implies \frac{-2}{x^2-2x}=3 \implies 3x^2-6x=-2 \implies 3x^2-6x+2=0 \quad \textbf{[1 mark]}$$

Applying the quadratic formula $(a=3,\ b=-6,\ c=2)$:

$$x=\frac{-(-6)\pm\sqrt{(-6)^2-4(3)(2)}}{2(3)}=\frac{6\pm\sqrt{36-24}}{6}=\frac{6\pm\sqrt{12}}{6}=\frac{6\pm 2\sqrt{3}}{6}=\frac{3\pm\sqrt{3}}{3}$$

Both values are non-zero and $\neq 2$. Hence, $\boxed{x=\dfrac{3+\sqrt{3}}{3} \text{ and } x=\dfrac{3-\sqrt{3}}{3}}$. **[1 mark]**

Q8. Find the discriminant of the quadratic equation $3x^2-2x+\dfrac{1}{3}=0$, and hence describe the nature of its roots. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $3x^2-2x+\dfrac{1}{3}=0$ with $ax^2+bx+c=0$, we have $a=3$, $b=-2$, and $c=\dfrac{1}{3}$.

Formula Used: Discriminant $D=b^2-4ac$

Computing D:

$$D=(-2)^2-4(3)\left(\frac{1}{3}\right)=4-4=\boxed{0} \quad \textbf{[1 mark]}$$

Since $D=0$, the equation has **two real and equal roots** (given by $x=-\dfrac{b}{2a}=\dfrac{2}{6}=\dfrac{1}{3},\ \dfrac{1}{3}$).

Hence, $\boxed{D=0;\ \text{two real and equal roots } \left(\frac{1}{3},\ \frac{1}{3}\right)}$. **[1 mark]**

Q9. Solve the following quadratic equation for x by factorization: $\sqrt{2}x^2 + 9x + 4\sqrt{2} = 0$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $\sqrt{2}x^2 + 9x + 4\sqrt{2} = 0$ with $ax^2 + bx + c = 0$, we have $a = \sqrt{2}$, $b = 9$, and $c = 4\sqrt{2}$.

Formula Used: $ac = (\sqrt{2})(4\sqrt{2}) = 4 \times 2 = 8, \qquad b = 9 = 8 + 1$

Splitting the middle term as $9x = 8x + x$ (noting $8 = 4\sqrt{2} \cdot \sqrt{2}$):

$$\sqrt{2}x^2 + 8x + x + 4\sqrt{2} = 0$$
$$\sqrt{2}x(x + 4\sqrt{2}) + 1(x + 4\sqrt{2}) = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$
$$(x + 4\sqrt{2})(\sqrt{2}x + 1) = 0$$

Equating each factor to zero:

$$x + 4\sqrt{2} = 0 \implies x = -4\sqrt{2} \qquad \text{or} \qquad \sqrt{2}x + 1 = 0 \implies x = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$$

Hence, the roots are $\boxed{x = -4\sqrt{2} \text{ and } x = -\frac{\sqrt{2}}{2}}$. **[1$\frac{1}{2}$ marks]**

Q10. Solve the following algebraic equation for x: $\dfrac{2x}{x-3} + \dfrac{1}{2x+3} + \dfrac{3x+9}{(x-3)(2x+3)} = 0, \quad x \neq 3, -\dfrac{3}{2}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Taking the common denominator $(x-3)(2x+3)$ across all three terms on the LHS:

Formula Used: $\dfrac{2x(2x+3) + 1(x-3) + (3x+9)}{(x-3)(2x+3)} = 0$

Since $x \neq 3$ and $x \neq -\dfrac{3}{2}$, $(x-3)(2x+3) \neq 0$, so multiplying both sides by $(x-3)(2x+3)$:

$$2x(2x+3) + (x-3) + (3x+9) = 0$$
$$4x^2 + 6x + x - 3 + 3x + 9 = 0 \implies 4x^2 + 10x + 6 = 0 \implies 2x^2 + 5x + 3 = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Factorising ($2 \times 3 = 6$; $2 + 3 = 5$):

$$2x^2 + 2x + 3x + 3 = 0 \implies 2x(x+1) + 3(x+1) = 0 \implies (x+1)(2x+3) = 0 \quad \textbf{[1 mark]}$$

Therefore, $x = -1$ or $x = -\dfrac{3}{2}$.

Domain Check: Since $x \neq -\dfrac{3}{2}$ is explicitly excluded (as $2x+3 = 0$ causes division by zero in $\frac{1}{2x+3}$), we **reject** $x = -\dfrac{3}{2}$.

Hence, the unique valid solution is $\boxed{x = -1}$. **[$\frac{1}{2}$ mark]**

Q11. Solve the following algebraic equation for x: $\dfrac{1}{x+1} + \dfrac{2}{x+2} = \dfrac{4}{x+4}, \quad x \neq -1, -2, -4.$ **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Combining the LHS terms over the common denominator $(x+1)(x+2)$:

Formula Used: $\dfrac{(x+2)+2(x+1)}{(x+1)(x+2)} = \dfrac{4}{x+4}, \qquad x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$\frac{x+2+2x+2}{x^2+3x+2} = \frac{4}{x+4} \implies \frac{3x+4}{x^2+3x+2} = \frac{4}{x+4} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Cross-multiplying:

$$(3x+4)(x+4) = 4(x^2+3x+2)$$
$$3x^2+16x+16 = 4x^2+12x+8 \implies x^2-4x-8=0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Applying the quadratic formula ($a=1$, $b=-4$, $c=-8$):

$$x = \frac{-(-4) \pm \sqrt{(-4)^2-4(1)(-8)}}{2(1)} = \frac{4 \pm \sqrt{16+32}}{2} = \frac{4 \pm \sqrt{48}}{2} = \frac{4 \pm 4\sqrt{3}}{2} = 2 \pm 2\sqrt{3}$$

Both $2+2\sqrt{3}$ and $2-2\sqrt{3}$ satisfy $x \neq -1, -2, -4$.

Hence, the solutions are $\boxed{x = 2+2\sqrt{3} \text{ and } x = 2-2\sqrt{3}}$. **[2 marks]**

Q12. A shopkeeper buys a number of books for Rs. 80. If he had bought 4 more books for the same total expenditure, each book would have cost Rs. 1 less. Determine how many books he originally bought and the original cost of each book. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the original number of books bought be x ($x \in \mathbb{N}$).

Total expenditure = Rs. 80, so original cost per book = Rs. $\dfrac{80}{x}$, and new cost per book for $(x+4)$ books = Rs. $\dfrac{80}{x+4}$. **[1 mark]**

Formula Used: Cost per book $= \dfrac{\text{Total Amount}}{\text{Number of books}}, \qquad \dfrac{80}{x} - \dfrac{80}{x+4} = 1$

Setting up the equation:

$$\frac{80}{x} - \frac{80}{x+4} = 1$$
$$80\left[\frac{(x+4)-x}{x(x+4)}\right] = 1 \implies \frac{320}{x^2+4x} = 1 \implies x^2+4x-320=0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Factorising ($20 \times (-16) = -320$ and $20-16=4$):

$$x^2+20x-16x-320=0 \implies x(x+20)-16(x+20)=0 \implies (x+20)(x-16)=0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Thus $x=-20$ or $x=16$. Since the number of books cannot be negative ($x>0$), we reject $x=-20$.

Therefore, original number of books $= 16$, and original cost per book $= \dfrac{80}{16} =$ Rs. 5. (Check: $\frac{80}{16} - \frac{80}{20} = 5-4 =$ Rs. 1.)

Hence, $\boxed{\text{Books originally bought} = 16,\ \text{Original cost per book} = \text{Rs. } 5}$. **[1 mark]**

Q13. Case Study: School Educational Excursion & Group Budget Dynamics

The student council of a senior secondary school planned a leadership excursion to a science exploration centre with a fixed bus charter contract of Rs. 2000, to be shared equally among all participating students. On the morning of the departure, 5 students fell ill and could not join. Consequently, the remaining students each had to contribute Rs. 20 more to cover the total bus charter expense. [4]

(i) If x denotes the original planned number of participating students, write expressions for the original and revised individual contributions. **[1]**

(ii) Formulate the quadratic equation in x in standard form representing the financial situation. **[1]**

(iii) Determine the original number of students planned (x) and the number of students who actually attended the excursion. **OR** Calculate the exact amount each attending student paid and the extra burden incurred per student due to the 5 cancellations. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

We know the equal-sharing cost relation:

Formula Used: Contribution per student $= \dfrac{\text{Total Bus Charter Cost}}{\text{Number of students}}$

(i) Contribution Expressions: Total cost = Rs. 2000, planned students $= x$ ($x > 5$), attending students $= x - 5$.

$$\boxed{\text{Original contribution} = \text{Rs. } \frac{2000}{x}} \quad \text{and} \quad \boxed{\text{Revised contribution} = \text{Rs. } \frac{2000}{x-5}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Form: Each remaining student paid Rs. 20 more:

$$\frac{2000}{x-5} - \frac{2000}{x} = 20$$

$$100\left[\frac{x-(x-5)}{x(x-5)}\right] = 1 \quad \text{(dividing by 20)}$$

$$\frac{500}{x^2-5x} = 1 \implies \boxed{x^2 - 5x - 500 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Planned & Attending Students): Factorising ($-25 \times 20 = -500$ and $-25 + 20 = -5$):

$$x^2 - 25x + 20x - 500 = 0 \implies (x-25)(x+20) = 0 \implies x = 25 \text{ or } x = -20$$

Since number of students cannot be negative ($x > 5$), we reject $x = -20$.
Thus originally planned students = 25, and students who actually attended = $25 - 5 = 20$.
Hence, $\boxed{\text{Planned} = 25 \text{ students, Attended} = 20 \text{ students}}$. **[2 marks]**

OR (Alternative Part — Contribution Paid & Extra Burden): Using $x = 25$ and $(x-5) = 20$ attending students:

$$\text{Amount each attending student paid} = \frac{2000}{20} = \text{Rs. } 100$$

$$\text{Original planned share} = \frac{2000}{25} = \text{Rs. } 80$$

Extra burden per student = $100 - 80 = $ Rs. 20.
Hence, $\boxed{\text{Each attending student paid} = \text{Rs. } 100, \text{ Extra burden} = \text{Rs. } 20 \text{ per student}}$. **[2 marks]**

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★